

GUJARAT TECHNOLOGICAL UNIVERSITY

Diploma Engineering – SEMESTER – 1 (NEW) – EXAMINATION – Summer-2026

Subject Code: DI01000021

Date: 16-06-2026

Subject Name: Mathematics-I

Time: 10:30 AM TO 01:00 PM

Total Marks: 70

Instructions:

1. Attempt all questions.
2. Make Suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Use of simple calculators and non-programmable scientific calculators are permitted.
5. English version is authentic.

Q.1 Fill in the blanks/MCQs using appropriate choice from the given options. 14

(યોગ્ય વિકલ્પ પસંદ કરી ખાલી જગ્યા પૂરો/ બહુવિકલ્પ પ્રશ્નોનાં જવાબ આપો)

1. $\begin{vmatrix} -1 & 3 \\ -4 & 6 \end{vmatrix} = \underline{\hspace{2cm}}$

- (A) - 18 (B) 6 (C) - 6 (D) 18

૧. $\begin{vmatrix} -1 & 3 \\ -4 & 6 \end{vmatrix} = \underline{\hspace{2cm}}$

- (અ) -18 (બ) 6 (ક) -6 (ડ) 18

2. If $f(x) = \log_2 x$, then $f(4) = \underline{\hspace{2cm}}$

- (A) 0 (B) 1 (C) 2 (D) 4

૨. જો $f(x) = \log_2 x$ તો $f(4) = \underline{\hspace{2cm}}$

- (અ) 0 (બ) 1 (ક) 2 (ડ) 4

3. If $\log_x \left(\frac{9}{16} \right) = -2$ then $x = \underline{\hspace{2cm}}$

- (A) $\frac{3}{4}$ (B) $\frac{4}{3}$ (C) $\frac{81}{256}$ (D) $\frac{256}{81}$

૩. જો $\log_x \left(\frac{9}{16} \right) = -2$ તો $x = \underline{\hspace{2cm}}$

- (અ) $\frac{3}{4}$ (બ) $\frac{4}{3}$ (ક) $\frac{81}{256}$ (ડ) $\frac{256}{81}$

4. $\begin{vmatrix} \log_{21} 3 & -1 \\ \log_{21} 7 & 1 \end{vmatrix} = \underline{\hspace{2cm}}$
- (A) 1 (B) -1 (C) 0 (D) 2
8. $\begin{vmatrix} \log_{21} 3 & -1 \\ \log_{21} 7 & 1 \end{vmatrix} = \underline{\hspace{2cm}}$
- (A) 1 (B) -1 (C) 0 (D) 2
5. $216^\circ = \underline{\hspace{2cm}}$ Radian
- (A) $\frac{5\pi}{6}$ (B) $\frac{5\pi}{3}$ (C) $\frac{6\pi}{5}$ (D) $\frac{3\pi}{5}$
4. $216^\circ = \underline{\hspace{2cm}}$ રેડિયન
- (A) $\frac{5\pi}{6}$ (B) $\frac{5\pi}{3}$ (C) $\frac{6\pi}{5}$ (D) $\frac{3\pi}{5}$
6. $\tan^{-1}x + \cot^{-1}x = \underline{\hspace{2cm}}. x \in R$
- (A) 1 (B) 0 (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{6}$
5. $\tan^{-1}x + \cot^{-1}x = \underline{\hspace{2cm}}. x \in R$
- (A) 1 (B) 0 (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{6}$
7. Principal period of $2\sin\theta \cdot \cos\theta = \underline{\hspace{2cm}}$
- (A) π (B) 2π (C) 0 (D) $\frac{\pi}{2}$
9. $2\sin\theta \cdot \cos\theta$ ની મુખ્ય અવર્તકાલ = $\underline{\hspace{2cm}}$
- (A) π (B) 2π (C) 0 (D) $\frac{\pi}{2}$
8. If $\vec{a} = (2, 1, -1)$ and $\vec{b} = (1, -3, 2)$ then $|2\vec{a} - 3\vec{b}| = \underline{\hspace{2cm}}$
- (A) $\sqrt{186}$ (B) $\sqrt{196}$ (C) $\sqrt{83}$ (D) 21
6. જો $\vec{a} = (2, 1, -1)$ અને $\vec{b} = (1, -3, 2)$ તો $|2\vec{a} - 3\vec{b}| = \underline{\hspace{2cm}}$
- (A) $\sqrt{186}$ (B) $\sqrt{196}$ (C) $\sqrt{83}$ (D) 21
9. If $(1, -2, 3) \cdot (4, 5, m) = 0$ $m = \underline{\hspace{2cm}}$
- (A) 2 (B) -2 (C) 3 (D) 0

9. $(1, -2, 3) \cdot (4, 5, m) = 0$ and $m =$ _____
- (A) 2 (B) -2 (C) 3 (D) 0
10. centre of the circle $x^2 + y^2 - 2x + 4y + 1 = 0$
- (A) (1, 2) (B) (1, -2) (C) (-1, -2) (D) (-1, 2)
11. circle $x^2 + y^2 - 2x + 4y + 1 = 0$ and $3x =$ _____
- (A) (1, 2) (B) (1, -2) (C) (-1, -2) (D) (-1, 2)
12. slope of the line $l: 2x - 5y + 3 = 0$ is _____
- (A) $\frac{2}{5}$ (B) $-\frac{2}{5}$ (C) $\frac{5}{2}$ (D) $-\frac{5}{2}$
13. line $2x - 5y + 3 = 0$ and $3x =$ _____
- (A) $\frac{2}{5}$ (B) $-\frac{2}{5}$ (C) $\frac{5}{2}$ (D) $-\frac{5}{2}$
14. $\lim_{x \rightarrow 2} \frac{x^5 - 32}{x - 2} =$ _____
- (A) 80 (B) 160 (C) 40 (D) 0
15. $\lim_{x \rightarrow 2} \frac{x^5 - 32}{x - 2} =$ _____
- (A) 80 (B) 160 (C) 40 (D) 0
16. $\lim_{\theta \rightarrow 0} \frac{\tan^2(\frac{\theta}{2})}{\theta^2} =$ _____
- (A) 4 (B) $\frac{1}{4}$ (C) 2 (D) $\frac{1}{2}$
17. $\lim_{\theta \rightarrow 0} \frac{\tan^2(\frac{\theta}{2})}{\theta^2} =$ _____
- (A) 4 (B) $\frac{1}{4}$ (C) 2 (D) $\frac{1}{2}$
18. $\lim_{h \rightarrow 0} \frac{a^h - 1}{h} =$ _____
- (A) $\log_e a$ (B) e (C) a (D) 0

૧૪. $\lim_{h \rightarrow 0} \frac{a^h - 1}{h} = \underline{\hspace{2cm}}$

(અ) $\log_e a$

(બ) e

(ક) a

(ડ) 0

Q.2 A Attempt any two (કોઇપણ બે ના જવાબ આપો)

6

1. If $\begin{vmatrix} x & 4 & 4 \\ 4 & x & 4 \\ 4 & 4 & 0 \end{vmatrix} = 0$ then find value of

૧. જો $\begin{vmatrix} x & 4 & 4 \\ 4 & x & 4 \\ 4 & 4 & 0 \end{vmatrix} = 0$ તો x ની કિંમત શોધો

2. Prove that $\frac{1}{\log_{24} 12} + \frac{1}{\log_8 12} + \frac{1}{\log_9 12} = 3$

૨. સાબિત કરો કે $\frac{1}{\log_{24} 12} + \frac{1}{\log_8 12} + \frac{1}{\log_9 12} = 3$

3. Prove that $\tan 35^\circ + \tan 10^\circ + \tan 35^\circ \cdot \tan 10^\circ = 1$

૩. સાબિત કરો કે $\tan 35^\circ + \tan 10^\circ + \tan 35^\circ \cdot \tan 10^\circ = 1$

Q.2 B Attempt any two (કોઇપણ બે ના જવાબ આપો)

8

1. If $f(x) = \frac{1-x}{1+x}$ then prove that (i) $f(x) + f\left(\frac{1}{x}\right) = 0$

(ii) $f(x) - f\left(\frac{1}{x}\right) = 2f(x)$

૧. જો $f(x) = \frac{1-x}{1+x}$ તો સાબિત કરો કે (i) $f(x) + f\left(\frac{1}{x}\right) = 0$

(ii) $f(x) - f\left(\frac{1}{x}\right) = 2f(x)$

2. If $\log(x-y) = \log 2 + \frac{1}{2} \log x + \frac{1}{2} \log y$ then prove that

(i) $x^2 + y^2 = 6xy$

(ii) $\frac{x}{y} + \frac{y}{x} = 6$

૨. જો $\log(x-y) = \log 2 + \frac{1}{2} \log x + \frac{1}{2} \log y$ તો સાબિત કરો કે

(i) $x^2 + y^2 = 6xy$

(ii) $\frac{x}{y} + \frac{y}{x} = 6$

3. If $\frac{\log_{10}x}{b-c} = \frac{\log_{10}y}{c-a} = \frac{\log_{10}z}{a-b}$ then prove that (i) $xyz = 1$

(ii) $x^a \cdot y^b \cdot z^c = 1$

3. જો $\frac{\log_{10}x}{b-c} = \frac{\log_{10}y}{c-a} = \frac{\log_{10}z}{a-b}$ તો સાબિત કરો કે (i) $xyz = 1$

(ii) $x^a \cdot y^b \cdot z^c = 1$

Q.3 A Attempt any two (કોઇપણ બે ના જવાબ આપો)

6

1. Prove that $\frac{\sin(\frac{\pi}{2}+\theta)}{\cos(2\pi-\theta)} + \frac{\tan(\pi+\theta)}{\cot(\frac{\pi}{2}-\theta)} + \frac{\sec(\frac{3\pi}{2}+\theta)}{\operatorname{cosec}(\pi-\theta)} = 3$

૧. સાબિત કરો કે $\frac{\sin(\frac{\pi}{2}+\theta)}{\cos(2\pi-\theta)} + \frac{\tan(\pi+\theta)}{\cot(\frac{\pi}{2}-\theta)} + \frac{\sec(\frac{3\pi}{2}+\theta)}{\operatorname{cosec}(\pi-\theta)} = 3$

2.. Prove that $\frac{\cos\theta + \cos3\theta + \cos5\theta}{\sin\theta + \sin3\theta + \sin5\theta} = \cot3\theta$

૨. સાબિત કરો કે $\frac{\cos\theta + \cos3\theta + \cos5\theta}{\sin\theta + \sin3\theta + \sin5\theta} = \cot3\theta$

3. Prove that lines $l_1: 3x - 2y + 5 = 0$ and $l_2: 2x + 3y - 7 = 0$ are perpendicular to each other

3. સાબિત કરો કે રેખા $l_1: 3x - 2y + 5 = 0$ અને $l_2: 2x + 3y - 7 = 0$ એકબીજાને લંબ છે

Q.3 B Attempt any two (કોઇપણ બે ના જવાબ આપો)

8

1. Prove that $8\cos(20^\circ) \cos(40^\circ) \cos(80^\circ) = 1$

૧. સાબિત કરો કે $8\cos(20^\circ) \cos(40^\circ) \cos(80^\circ) = 1$

2. draw the graph $y = \cos x, -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

૨. $y = \cos x, -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ નો આલેખ દોરો

૩. If $\vec{a} = 2\vec{i} - \vec{j}, \vec{b} = \vec{i} + 3\vec{j} - 2\vec{k}$ then find $|(\vec{a} + \vec{b}) \times (\vec{a} - \vec{b})|$

૩. જો $\vec{a} = 2\vec{i} - \vec{j}, \vec{b} = \vec{i} + 3\vec{j} - 2\vec{k}$ હોઈતો $|(\vec{a} + \vec{b}) \times (\vec{a} - \vec{b})|$ શોધો

Q.4 A Attempt any two (કોઇપણ બે ના જવાબ આપો)

6

1. If $\vec{a} = \vec{i} + 2\vec{j} + \vec{k}, \vec{b} = 2\vec{i} - 3\vec{j} + \vec{k}$ and $\vec{c} = -2\vec{i} - \vec{j} + 5\vec{k}$ then find

$$|2\vec{a} + 3\vec{b} - \vec{c}|$$

૧. જો $\vec{a} = \vec{i} + 2\vec{j} + \vec{k}, \vec{b} = 2\vec{i} - 3\vec{j} + \vec{k}$ અને $\vec{c} = -2\vec{i} - \vec{j} + 5\vec{k}$ હોઈ તો

$$|2\vec{a} + 3\vec{b} - \vec{c}|$$
 શોધો

2. For what value of P, the vectors $2\vec{i} + 3\vec{j} - 5\vec{k}$ and $P\vec{i} + \vec{j} + 3\vec{k}$ are perpendicular to each other

૨. P ની કઈ કિંમત માટે સદીશ $2\vec{i} + 3\vec{j} - 5\vec{k}$ અને $P\vec{i} + \vec{j} + 3\vec{k}$ એકબીજાને લંબ થશે

3. Prove that the points $(1,0), (-1,2), (0,1)$ are collinear

૩. સાબિત કરો કે બિંદુઓ $(1,0), (-1,2), (0,1)$ સમરેખ છે.

Q.4 B Attempt any two (કોઇપણ બે ના જવાબ આપો)

8

1. A particle moves from $(-1,2,1)$ to $(2,3,-1)$ under the effect of the constant force $(1,2,1)$ and $(2,-1,0)$, Find the work done

૧. એક કણ પર $(1,2,1)$ અને $(2,-1,0)$ બળો લાગે છે. આ બળની અસર હેઠળ કણ બિંદુ $(-1,2,1)$ થી $(2,3,-1)$ સુધી જાય છે. તો થયેલું કાર્ય શોધો.

2. Prove that the angle between the vectors $(1,1,-1)$ and $(2,-2,1)$ is $\sin^{-1}\left(\sqrt{\frac{26}{27}}\right)$

૨. સાબિત કરો કે સદીશ $(1,1,-1)$ અને $(2,-2,1)$ વચ્ચેનો ખૂણો $\sin^{-1}\left(\sqrt{\frac{26}{27}}\right)$ છે

3. Evaluate: $\lim_{x \rightarrow 2} \frac{x^3 - 3x^2 + 5x - 6}{x^3 - 8}$
3. કિમત મેળવો : $\lim_{x \rightarrow 2} \frac{x^3 - 3x^2 + 5x - 6}{x^3 - 8}$

Q.5 A Attempt any two (કોઇપણ બે ના જવાબ આપો)

6

1. Evaluate: $\lim_{x \rightarrow 0} \frac{\sqrt{25+x} - 5}{x}$
૧. કિમત મેળવો : $\lim_{x \rightarrow 0} \frac{\sqrt{25+x} - 5}{x}$
2. Evaluate: $\lim_{n \rightarrow \infty} (\sqrt{n^2 + n + 1} - n)$
૨. કિમત મેળવો : $\lim_{n \rightarrow \infty} (\sqrt{n^2 + n + 1} - n)$
3. Evaluate: $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$
3. કિમત મેળવો : $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$

Q.5 B Attempt any two (કોઇપણ બે ના જવાબ આપો)

8

1. Find the equation of the circle having centre (3,-1) and passing through (-1,2)
૧. કેન્દ્ર (૩,-૧) અને (-૧,૨) માંથી પસાર થતાં વર્તુળનું સમીકરણ શોધો
2. Find the equation of a line passing through the point (-6,2) and parallel to the line joining the points (-3,1) and (-5,2)
૨. (-૬,૨) માંથી પસાર થતી તેમજ (-૩,૧) અને (-૫,૨) ને જોડતી રેખાને સંમાતર રેખાનું સમીકરણ શોધો
3. Find the equation of tangent and normal to the circle

$$x^2 + y^2 - 4x + 2y + 3 = 0 \text{ at the point } (-1,2)$$

૩. વર્તુળ $x^2 + y^2 - 4x + 2y + 3 = 0$ નું બિંદુ $(-1,2)$ સ્પર્શક અને અભિલંબનું સમીકરણ મેળવો
